

SHAPEIC: DESIGN-SPACE EXPLORATION

Hierarchical LUT-based exploration from electrical models to layout-aware design

Daniel Arevalos

AEMY Group

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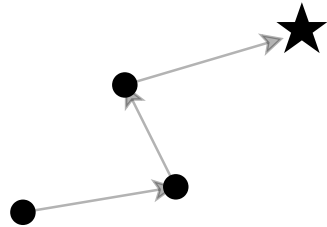
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Analog Sizing: Optimize or Explore ?

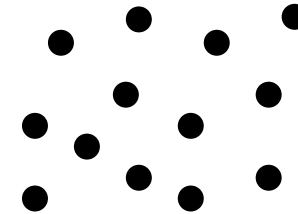
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Analog Sizing: Optimize or Explore ?

► Optimization-driven

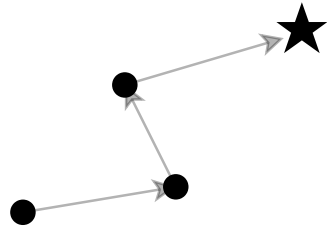


► Design-space exploration



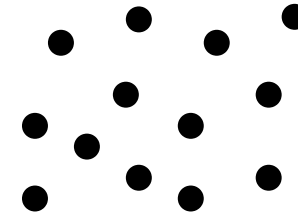
Analog Sizing: Optimize or Explore ?

► Optimization-driven



- ✓ Efficient search
- ✓ Many algorithms available
- ✗ Partial View of design space
- ✗ May converge to local optima

► Design-space exploration



- ✓ Design insight
- ✓ Alternative solutions
- ✗ Expensive to build
- ✗ Scales poorly without abstraction

ShapelC: Building the Design Space



ShapelC: Design-Space Exploration – Hierarchical LUT-based exploration from electrical models to layout-aware design

Daniel Arevalos – AEMY Group

Layout-Aware Exploration

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Current Status & What is Next ?

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